Seminar Quiz 4

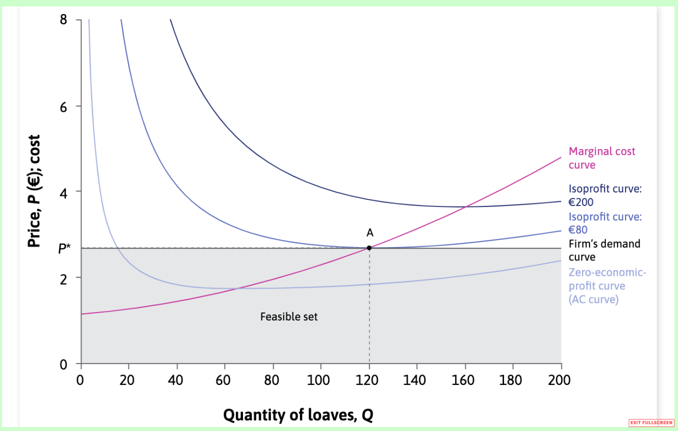
The Economy

Sciences Po Paris - Reims

15 minutes. No extra papers. You are free to use calculators.

Exercise 1

1. Draw the demand and MC curves for a **single** firm in a competitive market for rings. Show a hypothetical point at which the firm will trade, no numbers needed. (2 points + 1 point for good labeling)



- MC will shift **up** (remember up for an increase in prices, right for quantities) when the price of gold increases. With the same market price, the new equilibrium **A'** will be at a lower quantity. $\rightarrow$ higher marginal costs decrease quantity sold

2. In the same graph (or a copy of it), show what would happen if the price of gold went up. Assume that gold is an input variable for rings. (2 points)

Exercise 2

Consider a monopoly selling phones. For the point where trade happens between the monopolist and the consumers, we know the following:

$$P_{mono} = 200$$
$$Q_{mono} = 50$$
$$MR = 100$$

1. Given this information, calculate the slope of the isoprofit curve at the point where trade happens. (3 points)

The monopoly sets the price where $MC = MR$. Hence the slope of the isoprofit formula can be rewritten as:

$$-\frac{P - MC}{Q} = -\frac{P - MR}{Q} = -\frac{200 - 100}{50} = -2$$

At the point of trade, the slope of the isoprofit curve is equal to the slope of the demand line due to $MRS = MRT$. If the demand has a slope of -2 then its elasticity is $\frac{1}{2} \rightarrow$ inelastic.

2. Assuming linearity for this firm's demand formula, which of the following statements is true for this market? (2 point)

A) The demand curve is *elastic* so the DWL is *bigger* than if the demand curve were inelastic.

B) The demand curve is *inelastic* so the DWL is *bigger* than if the demand curve were elastic.

C) The demand curve is *elastic* so the DWL is *smaller* than if the demand curve were inelastic.

D) The demand curve is *inelastic* so the DWL is *smaller* than if the demand curve were elastic.